Analysis Report: AI-Driven Feature File Generation and Endpoint Coverage Identifier

Title: AI-Driven Feature File Generation

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1. Problem Statement

In the current software testing process, the creation of feature files and the identification of API endpoints that have already been tested are performed manually. This approach is time-consuming, error-prone, and lacks consistency across test artifacts. As the number of endpoints grows, it becomes increasingly difficult to track test coverage and ensure that all critical functionalities are adequately tested.

To improve efficiency, accuracy, and traceability in API testing, there is a need to **automate the generation of feature files** (used in behavior-driven development frameworks such as Cucumber) and to **programmatically identify the list of API endpoints that are already covered by existing tests**. This automation will streamline the test development process, reduce manual effort, and enhance visibility into test coverage across the system.

2. Executive Summary

This document evaluates AI-assisted approaches for generating **Cucumber test cases** using **Large Language Models (LLMs)**, specifically targeting **Python-based GenAI applications** with source code and API specifications (e.g., openapi.json and GitHub exports). The goal is to **automate Behavior-Driven Development (BDD)** processes at scale while handling real-world constraints like token limits and hallucinations.

3. Key Objective

To automate the creation of BDD-compliant Gherkin feature files for RESTful APIs using LLMs, reducing manual QA effort and ensuring alignment with documented and implemented behavior.

4.Scope

- **Automated Generation of BDD Test Cases:**
  The application automates creation of **Gherkin-style Cucumber feature files** for **C#-based APIs** by analyzing source code and OpenAPI specifications.
- **Input Sources:**
  - C# code repositories exported from GitHub, including controllers, services, and models typical in .NET projects.

    - OpenAPI (openapi.json) specifications describing RESTful APIs implemented in C#.

- **Core Functionalities:**
    - Parsing and semantic embedding of C# source code files using CodeBERT or similar embeddings trained for C# code.
    - Retrieval of relevant code chunks through vector search to handle token limitations of LLMs.
    - Prompt engineering that ensures generation of valid and comprehensive BDD scenarios, including success, validation errors, and edge cases.
    - Extraction of dynamic variables from responses, creation of data tables, and validation steps in Gherkin syntax.
- **Behavior-Driven Development (BDD) Compliance:**
  Generated feature files follow the Given-When-Then structure to bridge communication between developers, QA, and business stakeholders.
- **Scalability:**
  Supports medium to large-scale C# repositories by chunking code, embedding, and retrieval to optimize input size and maintain context.

Boundaries

- **Static Code and API Spec Analysis Only:**
  The application focuses on static analysis of C# source code and OpenAPI specs and does **not** perform runtime or dynamic analysis such as dependency injection behavior, runtime configuration, or middleware effects.
- **Dependent on Code Documentation Quality:**
  Effectiveness depends on well-documented C# codebases including XML doc comments and meaningful method names to improve semantic retrieval.
- **No Test Execution or Integration:**
  The system generates Cucumber feature files but does **not** run the tests or validate their execution in the target environment.
- **Token Limit and Context Constraints:**
  While RAG mitigates token limits, extremely large or complex codebases may still challenge context windows without additional chunking or orchestration.
- **Hallucination Risks:**
  Although minimized by retrieval-based context, hallucinations may occur during multi-step generation processes or when context linkage is incomplete.

- **Technology Scope:**
  Currently tailored for **C# REST API services** using OpenAPI specs; other API architectures or languages require further customization.
- **User Interaction Scope:**
  The tool operates as a backend or CLI service, with no native UI for manual test case modification or interactive refinement.

4. Theory

Large Language Models (LLMs):

LLMs like GPT-4 and CodeBERT are advanced AI models trained on extensive datasets of natural language and source code. They can generate structured outputs such as Cucumber test cases by understanding both code and API specifications in natural language and programming syntax.

Token Limit Constraints:

LLMs have a maximum token capacity (input + output), which limits processing very large inputs like full codebases or detailed API docs. Efficient handling of token limits is crucial to maintain context and output quality.

Retrieval-Augmented Generation (RAG):

RAG improves generation by combining semantic search with LLMs. Documents and code are split into chunks, embedded into vector representations using models like CodeBERT, and stored in a vector database. Upon a prompt, the system retrieves the most relevant chunks to feed the LLM, reducing input size while preserving context. This lowers hallucination risk and scales to large repositories.

LlamaIndex:

LlamaIndex (formerly known as GPT Index) is a data framework designed to connect LLMs with external knowledge sources such as code repositories or documentation. It manages the ingestion, indexing, and retrieval of documents, providing an interface for efficient retrieval in RAG workflows. Using LlamaIndex, the application can chunk, tag, and query C# source code files and OpenAPI specs effectively, facilitating precise context retrieval for test generation.

Amazon Bedrock:

Amazon Bedrock is a fully managed service that enables developers to build and scale generative AI applications using foundation models from leading AI providers without managing infrastructure. Bedrock can host LLMs that process the retrieved context and generate the Cucumber test cases at scale, integrating seamlessly with the retrieval layer built using LlamaIndex and embedding models like CodeBERT.

OpenAPI Specification:

The OpenAPI spec is a machine-readable definition of REST APIs, detailing endpoints, parameters, request/response schemas, and security. It guides the test generation by providing expected API behaviors and payload structures.

5. Proposed Solutions

Approach 1: Direct Input to LLM

Step 1: Export codebase from GitHub and obtain the openapi.json specification.
Step 2: Concatenate relevant code files and API spec into a single prompt.
Step 3: Feed the prompt directly into the LLM to generate test cases.

Pros:

- Simple and fast setup.
- Minimal processing or tooling required.

Cons:

- **Token Limit Breach**: Larger codebases exceed the LLM's context window.
- **Loss of Context**: Important connections between files may be omitted.
- **LLM Hallucinations**: May invent or misinterpret data due to lack of structure.

The flow diagram in figure 5a outlines a straightforward approach where the codebase and API specification are first exported from GitHub. Next, relevant files and the API spec are combined into a single prompt. Finally, this prompt is fed directly into the LLM to generate Cucumber test cases.

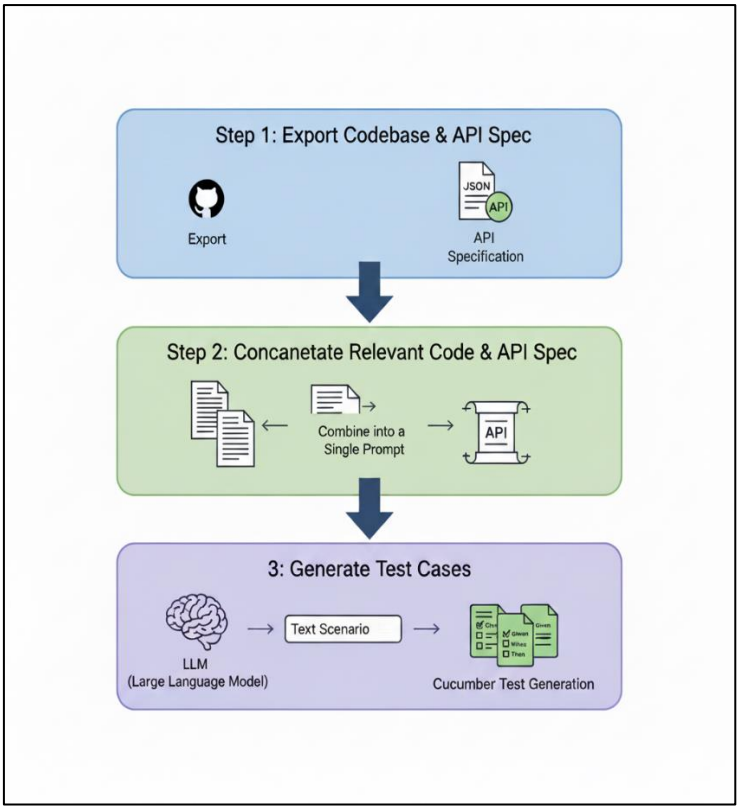

Figure 5a: Flow Chart for Pipeline Without RAG

Approach 2: RAG-Based Test Case Generation with LlamaIndex and CodeBERT

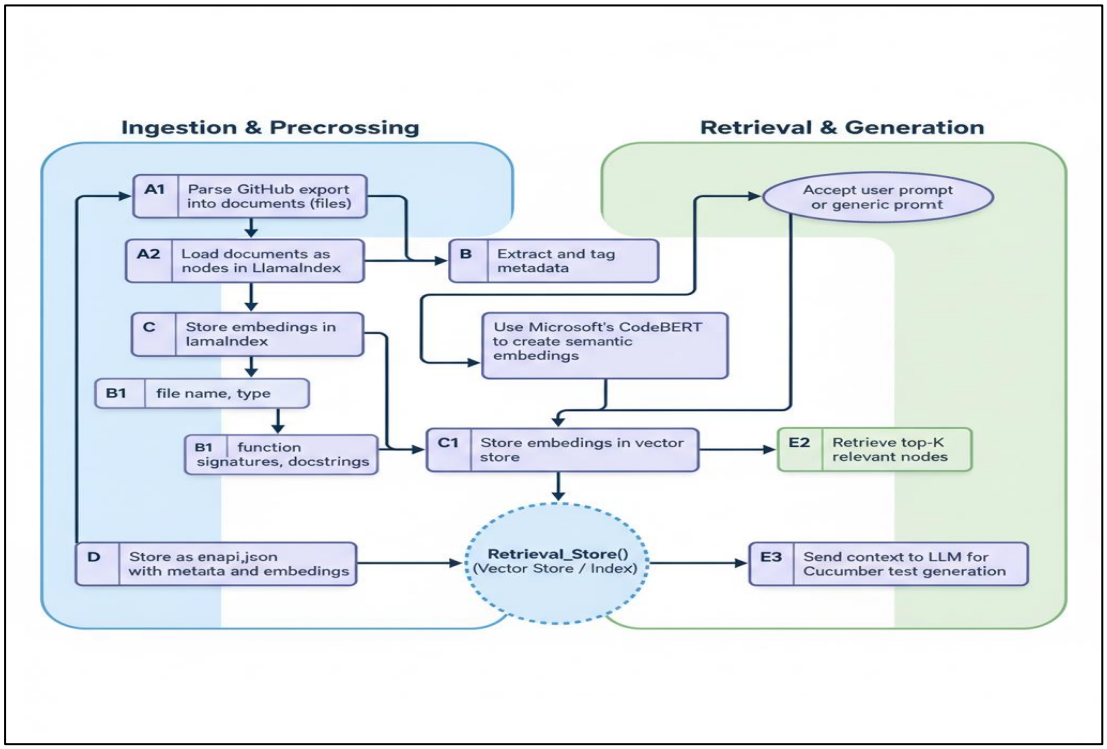

Figure 5b: Flow Chart for RAG Pipeline

Step 1: Parse GitHub export into documents (files) and load them as nodes in **LlamaIndex**.

Step 2: Extract and tag metadata such as file name, type (e.g., controller, service), function signatures, and docstrings.

Step 3: Use **Microsoft's CodeBERT** to create semantic embeddings for each node and store them in a vector store.

Step 4: Parse openapi.json and store as another node with its own metadata and embeddings.

Step 5: Use **Retrieval-Augmented Generation (RAG)**:

- Accept user prompt or generate a generic test case prompt.
- Retrieve top-K relevant nodes (e.g., endpoint handlers, service methods).
- Send retrieved context to LLM for Cucumber test case generation.

Pros:

- **Scalability**: Handles large codebases by chunking and embedding.
- **Context-Awareness**: Retrieves only relevant code to avoid hallucinations.
- **Extensible**: Can adapt to other documentation formats or specs.

Cons:

- **Complex Setup**: Requires additional tooling (LlamaIndex, vector DB).
- **Latency**: Retrieval and embedding add processing time.
- **Context Chain Hallucination**: When chaining multiple steps (e.g., feature + step definitions + glue code), LLM may hallucinate due to limited window size or mis linked context.

The figure 5b gives an overall overview of the flow of the python application to generate the feature file.

6. Comparative Analysis

Criteria	Approach 1 (Direct LLM)	Approach 2 (RAG + LlamaIndex + CodeBERT)
Setup Complexity	Low	Medium to High
Token Limit Handling	Poor	Good
Contextual Understanding	Weak	Strong
Scalability to Large Codebases	Low	High
Accuracy of Test Cases	Moderate	High
Hallucination Risk	High	Lower (but not zero)
Maintenance Overhead	Low	Moderate
Flexibility (adding new sources)	Low	High

7. Technologies Used

1. **Python:**

   Primary programming language used to implement the GenAI application, handle data processing, and orchestrate the test case generation pipeline.

2. **Pandas:**

   Python library used for data manipulation and analysis, especially for

processing tabular data such as API specifications, test case data, and intermediate outputs.

3. **Anthropic Claude 3.5 Sonnet Model (via Amazon Bedrock):**
   Accessed through Bedrock's API to perform natural language generation tasks, including creating Gherkin-style test cases from retrieved context.
4. **LlamaIndex (Python SDK):**
   Used for indexing and querying C# source code and OpenAPI documents, facilitating efficient retrieval of relevant code chunks.
5. **CodeBERT (Embedding Model):**
   Integrated into the Python pipeline to generate semantic embeddings of C# source files, enabling vector-based similarity search.
6. **Vector Database (e.g., FAISS):**
   Utilized to store and query embeddings for fast retrieval during the RAG process.
7. **JSON Parsing Libraries:**
   Used for reading and handling OpenAPI specification files.
8. **Git Tools:**
   For exporting and managing C# code repositories during data ingestion.

8. Limitations

1. **LLM Hallucinations during Chaining**:
   When attempting to generate entire feature files, step definitions, and glue code in a chained manner, hallucinations occur. This happens either due to:
   - Token window being exceeded during the chain.
   - Lack of fully connected context between retrieved chunks.
2. **Embedding Quality Sensitivity**:
   Poor embeddings (e.g., if code comments are absent or ambiguous) can lead to irrelevant chunk retrievals.

3. **Limited Support for Dynamic Runtime Context**:
   Static analysis (embeddings and metadata) may not capture runtime behaviours like dependency injection or dynamic routing.

4. **Vector Store Scaling**:
   As repositories grow, maintaining and querying the vector index becomes more resource intensive.

9. Recommendations

- **Use Approach 2 with prompt engineering constraints**, would generate better templates.
- **Chunk judiciously** — overly small or overly large chunks both reduce retrieval effectiveness.
- **Include function-level docstrings and API summaries** to improve embedding and retrieval quality.
- For more advanced chaining, consider integrating workflow **orchestration (e.g., LangGraph)** to manage token context more effectively across steps.

10. Result and Analysis

10.1 Result

The images below Fig 10.1 a and Fig 10.1 b show a direct comparison between a manually authored feature file and one generated automatically using the LLM. Also Fig 10.1 c show the data file generated Both are implement the same API "api/ItemSale/CheckoutCart".

```
14 removals                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
1 Feature: API POST Request using feature file for Checkout Cart in New Cart with Item
2
3   @post
4   Scenario Outline: Verify POST request for "<TestCaseName>"
5     Given User loads the base URL from config
6     And User sets up the User API request
7     And User load the request body from "<requestBodyPath>"
8     And User replace the variables in the request body "<VariablesToBeReplacedUsingExtractedVariables>" from example index "<ExampleIndex>"
9     When User sends a POST request to "<endpoint>"
10    And validate "<ValidatingResponseData>" from response body
11    Then the response status code should be <ExpectedStatusCode>
12    And the response time should be less than <ExpectedResponseTime> ms
13    And User logs the response body
14
15    Examples:
16      | TestCaseName | requestBodyPath | endpoint | VariablesToBeReplacedUsingExtractedVariables | ExampleIndex | ValidatingResponseData | ExpectedStatusCode | ExpectedResponseTime |
17      | Checkout Cart | src/test/resources/data/ItemSale/CheckoutCart/CheckoutCartData.json | /api/ItemSale/CheckoutCart | cartId | POSTEmptyCart0 | cart.totals.saletotalAmount:398.0,success:true | 200 | 2000 |
18
```

Fig 10.1 a: Manual Feature File output

```
1 Feature: API POST Request using feature file for Checkout Cart
2
3   @post
4   Scenario Outline: Verify POST request for "<TestCaseName>"
5     Given User loads the base URL from config
6     And User sets up the User API request
7     And User load the request body from "<requestBodyPath>"
8     And User replace the variables in the request body "<VariablesToBeReplacedUsingEnvironmentVariables>"
9     And User replace the variables in the request body "<VariablesToBeReplacedUsingExtractedVariables>" from example index "<ExampleIndex>"
10    When User sends a POST request to "<endpoint>"
11    And validate "<ValidatingResponseData>" from response body
12    Then the response status code should be <ExpectedStatusCode>
13    And the response time should be less than <ExpectedResponseTime> ms
14    And User logs the response body
15
16    Examples:
17      | TestCaseName | requestBodyPath | endpoint | VariablesToBeReplacedUsingEnvironmentVariables | VariablesToBeReplacedUsingExtractedVariables | ExampleIndex | ValidatingResponseData | ExpectedStatusCode | ExpectedResponseTime |
18      | Checkout Cart with Valid Payment | src/test/resources/data/ItemSale/CheckoutCart/CheckoutCartValidPayment.json | /api/ItemSale/CheckoutCart | cartId,paymentReferenceNumber | POSTEmptyCart0 | success:true,cart.cartState:Finished,cart.payments[0].paymentReferenceNumber:{{paymentReferenceNumber}} | 200 | 2000 |
19      | Checkout Cart without Payment | src/test/resources/data/ItemSale/CheckoutCart/CheckoutCartNoPayment.json | /api/ItemSale/CheckoutCart | cartId | | POSTEmptyCart1 | Type:CartNotReadyForCompletionFault,Message:Cart is not ready for completion | 400 | 2000 |
```

Fig 10.1 b: Generated Feature File output

```
2 removals                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
1 {
2   "cartItemIdentificator": 0,
3   "cartId": {
4     "id": "{{cartId}}",
5     "revision": 0
6   }
7 }

2 additions
1 {
2   "cartId": {
3     "id": "{{cartId}}",
4     "revision": 0
5   },
6   "cartItemIdentificator": {{cartItemIdentificator}}
7 }
```

Fig 10.1 c: Generated Data File output

10.2 Analysis

The bar chart shown in the figure 10.2 a compares the average feature file generation time per endpoint between manual and LLM-automated approaches. The manual process takes significantly longer, averaging over 20 minutes per

endpoint, whereas the LLM-automated method drastically reduces the time to just around 1 minute. This highlights the efficiency and time-saving potential of using Large Language Models to automate test case generation in software development.

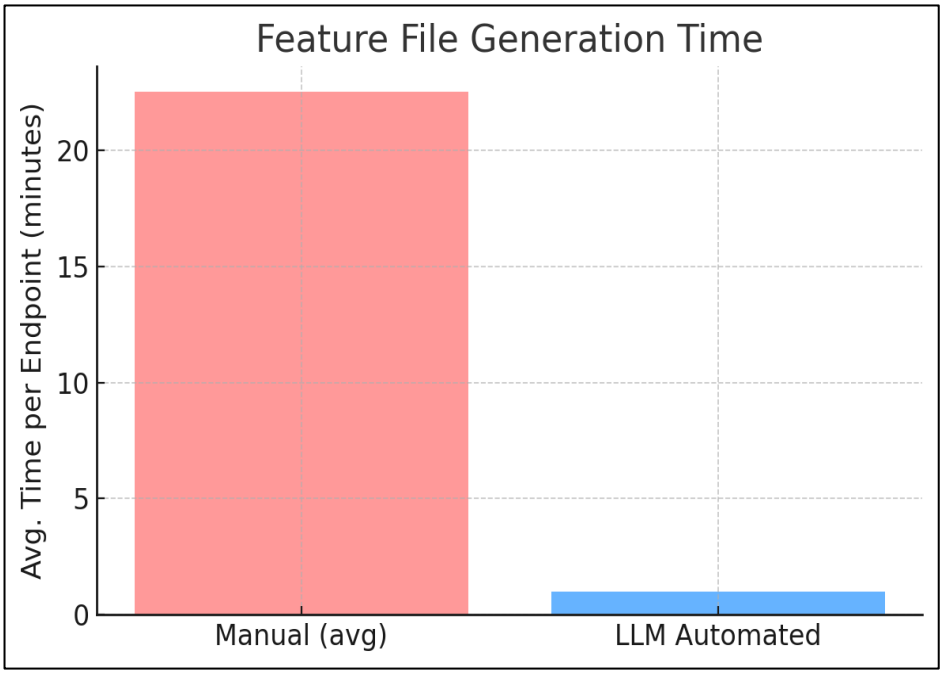

Fig 10.2 a: Feature File Generation Time

11. Conclusion

Both approaches offer value depending on the scale and complexity of the project:

- Approach 1 is viable for small projects or proof-of-concept testing, where the entire codebase and API spec can fit within token limits.
- Approach 2 is more robust for production use in medium to large repositories. The combination of metadata tagging, semantic embedding (CodeBERT), and RAG enables more accurate, relevant, and context-aware test case generation.
- The proposed solution would generate the feature file template in less time compared to manually creating it, but while generating the payload it might hallucinate.